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September 10, 2026

Editor-in-Chief
Computing
Springer

Dear Editor,

Please find enclosed our manuscript, “**ARMOR-FL: Anytime-Valid Robust Martingale-Based Online Reweighting for Federated Intrusion Detection**” (Manuscript ID 36ffcd3-31c3-4d49-8e4a-df0d92256555), which we submit for consideration for publication in *Computing*.

Federated learning lets distributed nodes train a shared model without pooling raw data, but it exposes the central aggregator to poisoned updates from compromised or malicious participants – a dependability problem that grows more acute, not less, as such systems are deployed across heterogeneous, non-identically-distributed networks of autonomous nodes. Byzantine-robust aggregation is the standard defense, and a recent line of work brings anytime-valid sequential testing to this problem, offering a formal, cumulative statistical guarantee on false-exclusion behavior that ordinary per-round heuristics cannot provide. In developing such a defense for federated network-intrusion detection, we identified and formally characterized a previously undiagnosed failure mode in this approach: an anytime-valid test that compares a client only against the population is mathematically guaranteed to eventually exclude any client whose behavior sits persistently off-center, so it cannot distinguish an honest node in a heterogeneous network from a persistent attacker – and the failure grows worse, not better, as the network becomes more homogeneous, the opposite of the intuitive expectation.

Our manuscript introduces ARMOR-FL, which corrects this failure with a self-referential shift test, a scale-floored population statistic, and a threshold-calibrated trust weight, and validates the fix on isolated synthetic reproductions and on three real intrusion-detection benchmarks (CICIDS2017, CICIDS2018, and CICIoT2023) under three attack types of increasing subtlety. We report our results plainly: ARMOR-FL does not outperform the simpler Krum baseline in aggregate, and we discuss at length why, and under which specific network conditions the reverse holds. We believe this combination – a formally grounded dependability mechanism for autonomic, self-protecting distributed systems, an honest diagnostic account of where it does and does not help, and a concrete deployment guideline grounded in measurable network properties – fits squarely within *Computing*’s scope on autonomic and dependable computing and internet computing, and will be of direct interest to its readership working on robust distributed and networked systems.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the manuscript and this submission. We have no conflicts of interest to disclose. The full implementation, experiment configuration grids, and reassembly-only dataset archives are released alongside the manuscript to support independent verification and extension.

Thank you for considering our manuscript. We look forward to your response.

Sincerely,

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